Análisis estadístico — exp-sweep-01

Experimento `exp-sweep-01` · snapshot `20261022T002233Z`

Generado 2026-10-29 21:49 UTC · dataset `wine_quality` · semillas 7, 42, 123, 999, 2025 · directorio de exportación `results\exports\exp-sweep-01`

CONFIGURACIONES

72

360 corridas

MEJOR F1 MEDIO

0.3034

tanh | adam | lr=0.01 | he | cw=balanced

MÁS RÁPIDA A LA MEJOR ÉPOCA

0.75 s

mínimo de las medias por configuración

REFERENCIA

grp-v1-7d12b4d9b658

tanh | adam | lr=0.01 | he | cw=balanced

> **Cómo leer este reporte.** Cada configuración se resume a través de sus semillas. La referencia es `grp-v1-7d12b4d9b658` (tanh | adam | lr=0.01 | he | cw=balanced), seleccionada porque: referencia empírica: mayor F1 macro medio (desempate: menor tiempo hasta la mejor época, menor desvío). Con 5 semillas por configuración, los tests inferenciales tienen baja potencia: tratá los p-valores como evidencia, no como prueba, y preferí los tamaños de efecto y los intervalos de confianza al decidir qué llevar a la siguiente etapa.

Observaciones clave

- Mejor F1 macro medio: **tanh | adam | lr=0.01 | he | cw=balanced** con 0.3034 ± 0.0328 (n=5).
- Mejor `class_weight=balanced` : tanh | adam | lr=0.01 | he | cw=balanced con 0.3034 (puesto 1).
- Mejor `class_weight=none` : tanh | adam | lr=0.001 | xavier | cw=none con 0.2714 (puesto 19).
- Menor tiempo medio hasta la mejor época: **relu | adam | lr=0.01 | he | cw=none** con 0.75 s, y puesto 41 en F1.
- 38 de 360 corridas válidas (10.6%) alcanzó el tope de épocas en su mejor época, por lo que su mejor modelo podría seguir mejorando.
- Mayor separación marginal: **Peso de clase** (Kruskal p=< 0.001), de balanced con 0.2486 a none con 0.2402.

> **Notas de calidad de datos**
>
> - class_weight divide la exportación en dos variantes experimentales; el reporte las muestra por separado y nunca las mezcla en silencio

1. Configuraciones de esta exportación

La tabla lista las 72 configuraciones con al menos una corrida. `n válidas/total` cuenta las corridas en `COMPLETED` o `COMPLETED_WITH_WARNINGS` sobre todas las corridas registradas. Las columnas de convergencia se derivan del historial por época: `best_epoch` es la época con menor pérdida de validación, `epochs_executed` es donde el early stopping terminó la corrida, y `time_to_best` es la suma acumulada de `epoch_time` hasta la mejor época.

Tabla 1a. Calidad por configuración, ordenada por F1 macro medio. El IC es el intervalo t del 95 % de la media entre semillas.

Puesto	Configuración	class_weight	n válidas/total	F1 medio	F1 desvío	IC 95 % de F1	Pérdida val.
1	**tanh | adam | lr=0.01 | he | cw=balanced**	balanced	5/5	0.3034	0.0328	[0.2626, 0.3441]	1.2376
2	**relu | sgd | lr=0.1 | xavier | cw=balanced**	balanced	5/5	0.2963	0.0260	[0.2640, 0.3285]	1.2367
3	**tanh | sgd | lr=0.1 | he | cw=balanced**	balanced	5/5	0.2953	0.0271	[0.2617, 0.3289]	1.2545
4	**tanh | adam | lr=0.001 | he | cw=balanced**	balanced	5/5	0.2869	0.0189	[0.2635, 0.3104]	1.2615
5	**tanh | adam | lr=0.01 | xavier | cw=balanced**	balanced	5/5	0.2854	0.0370	[0.2395, 0.3314]	1.2369
6	**relu | adam | lr=0.01 | he | cw=balanced**	balanced	5/5	0.2849	0.0389	[0.2366, 0.3332]	1.2640
7	**relu | adam | lr=0.001 | xavier | cw=balanced**	balanced	5/5	0.2844	0.0179	[0.2622, 0.3065]	1.2679
8	**relu | adam | lr=0.001 | he | cw=balanced**	balanced	5/5	0.2840	0.0097	[0.2720, 0.2960]	1.2664
9	**tanh | sgd | lr=0.01 | he | cw=balanced**	balanced	5/5	0.2839	0.0216	[0.2570, 0.3107]	1.2698
10	**tanh | adam | lr=0.001 | xavier | cw=balanced**	balanced	5/5	0.2830	0.0182	[0.2604, 0.3057]	1.2646
11	**tanh | sgd | lr=0.1 | xavier | cw=balanced**	balanced	5/5	0.2818	0.0169	[0.2608, 0.3027]	1.2180
12	**relu | sgd | lr=0.1 | he | cw=balanced**	balanced	5/5	0.2793	0.0201	[0.2543, 0.3043]	1.2587
13	**sigmoid | adam | lr=0.01 | xavier | cw=balanced**	balanced	5/5	0.2778	0.0206	[0.2523, 0.3033]	1.2018
14	**tanh | sgd | lr=0.01 | xavier | cw=balanced**	balanced	5/5	0.2770	0.0246	[0.2464, 0.3076]	1.2820
15	**sigmoid | sgd | lr=0.1 | xavier | cw=balanced**	balanced	5/5	0.2763	0.0264	[0.2434, 0.3091]	1.2905
16	**sigmoid | sgd | lr=0.1 | he | cw=balanced**	balanced	5/5	0.2758	0.0248	[0.2450, 0.3065]	1.2887
17	**sigmoid | adam | lr=0.01 | he | cw=balanced**	balanced	5/5	0.2751	0.0143	[0.2573, 0.2929]	1.2098
18	**relu | adam | lr=0.01 | xavier | cw=balanced**	balanced	5/5	0.2725	0.0281	[0.2376, 0.3074]	1.2596
19	**tanh | adam | lr=0.001 | xavier | cw=none**	none	5/5	0.2714	0.0214	[0.2448, 0.2980]	1.0459
20	**relu | sgd | lr=0.01 | xavier | cw=balanced**	balanced	5/5	0.2701	0.0119	[0.2553, 0.2849]	1.2722

Puesto	Configuración	class_weight	n válidas/total	F1 medio	F1 desvío	IC 95 % de F1	Pérdida val.
21	tanh | adam | lr=0.001 | he | cw=none	none	5/5	0.2678	0.0357	[0.2236, 0.3121]	1.0657
22	relu | sgd | lr=0.01 | he | cw=balanced	balanced	5/5	0.2657	0.0058	[0.2585, 0.2728]	1.2715
23	relu | sgd | lr=0.01 | he | cw=none	none	5/5	0.2649	0.0048	[0.2589, 0.2708]	1.0534
24	tanh | adam | lr=0.01 | he | cw=none	none	5/5	0.2646	0.0307	[0.2265, 0.3026]	1.0584
25	relu | sgd | lr=0.01 | xavier | cw=none	none	5/5	0.2642	0.0143	[0.2464, 0.2820]	1.0432
26	relu | sgd | lr=0.1 | xavier | cw=none	none	5/5	0.2640	0.0097	[0.2519, 0.2761]	1.0366
27	relu | adam | lr=0.001 | xavier | cw=none	none	5/5	0.2634	0.0081	[0.2534, 0.2735]	1.0426
28	relu | sgd | lr=0.1 | he | cw=none	none	5/5	0.2633	0.0111	[0.2495, 0.2770]	1.0420
29	sigmoid | sgd | lr=0.1 | xavier | cw=none	none	5/5	0.2630	0.0068	[0.2545, 0.2715]	1.0398
30	sigmoid | sgd | lr=0.1 | he | cw=none	none	5/5	0.2594	0.0065	[0.2513, 0.2675]	1.0443
31	tanh | sgd | lr=0.1 | xavier | cw=none	none	5/5	0.2594	0.0080	[0.2495, 0.2693]	1.0521
32	tanh | adam | lr=0.01 | xavier | cw=none	none	5/5	0.2590	0.0284	[0.2237, 0.2944]	1.0487
33	sigmoid | adam | lr=0.001 | he | cw=balanced	balanced	5/5	0.2583	0.0219	[0.2312, 0.2855]	1.3632
34	sigmoid | adam | lr=0.001 | xavier | cw=none	none	5/5	0.2568	0.0092	[0.2453, 0.2682]	1.0393
35	sigmoid | adam | lr=0.01 | he | cw=none	none	5/5	0.2565	0.0104	[0.2437, 0.2694]	1.0402
36	sigmoid | adam | lr=0.001 | xavier | cw=balanced	balanced	5/5	0.2560	0.0218	[0.2289, 0.2830]	1.3570
37	sigmoid | adam | lr=0.001 | he | cw=none	none	5/5	0.2548	0.0081	[0.2447, 0.2648]	1.0455
38	tanh | sgd | lr=0.1 | he | cw=none	none	5/5	0.2545	0.0153	[0.2355, 0.2735]	1.0719
39	relu | adam | lr=0.001 | he | cw=none	none	5/5	0.2545	0.0069	[0.2459, 0.2631]	1.0549
40	relu | adam | lr=0.01 | xavier | cw=none	none	5/5	0.2520	0.0172	[0.2306, 0.2733]	1.0487
41	relu | adam | lr=0.01 | he | cw=none	none	5/5	0.2513	0.0168	[0.2305, 0.2721]	1.0507
42	relu | sgd | lr=0.001 | xavier | cw=balanced	balanced	5/5	0.2503	0.0105	[0.2372, 0.2633]	1.3601
43	tanh | sgd | lr=0.01 | xavier | cw=none	none	5/5	0.2499	0.0122	[0.2347, 0.2650]	1.0580
44	tanh | sgd | lr=0.01 | he | cw=none	none	5/5	0.2481	0.0070	[0.2394, 0.2568]	1.0790

Puesto	Configuración	class_weight	n válidas/total	F1 medio	F1 desvío	IC 95 % de F1	Pérdida val.
45	**sigmoid | adam | lr=0.01 | xavier | cw=none**	none	5/5	0.2474	0.0102	[0.2348, 0.2600]	1.0390
46	**sigmoid | adam | lr=0.1 | he | cw=none**	none	5/5	0.2455	0.0157	[0.2260, 0.2650]	1.0820
47	**relu | sgd | lr=0.001 | xavier | cw=none**	none	5/5	0.2411	0.0114	[0.2270, 0.2553]	1.0754
48	**tanh | sgd | lr=0.001 | xavier | cw=none**	none	5/5	0.2405	0.0081	[0.2304, 0.2505]	1.0835
49	**relu | sgd | lr=0.001 | he | cw=balanced**	balanced	5/5	0.2402	0.0098	[0.2281, 0.2523]	1.3719
50	**relu | sgd | lr=0.001 | he | cw=none**	none	5/5	0.2398	0.0109	[0.2263, 0.2533]	1.0861
51	**tanh | sgd | lr=0.001 | he | cw=balanced**	balanced	5/5	0.2386	0.0164	[0.2182, 0.2589]	1.3942
52	**sigmoid | sgd | lr=0.01 | xavier | cw=balanced**	balanced	5/5	0.2378	0.0122	[0.2226, 0.2530]	1.4338
53	**tanh | sgd | lr=0.001 | xavier | cw=balanced**	balanced	5/5	0.2362	0.0078	[0.2265, 0.2459]	1.4011
54	**sigmoid | sgd | lr=0.01 | he | cw=balanced**	balanced	5/5	0.2327	0.0112	[0.2188, 0.2465]	1.4345
55	**sigmoid | adam | lr=0.1 | xavier | cw=balanced**	balanced	5/5	0.2323	0.0199	[0.2076, 0.2571]	1.2793
56	**tanh | sgd | lr=0.001 | he | cw=none**	none	5/5	0.2306	0.0089	[0.2195, 0.2417]	1.1001
57	**sigmoid | adam | lr=0.1 | xavier | cw=none**	none	5/5	0.2291	0.0112	[0.2152, 0.2429]	1.0797
58	**tanh | adam | lr=0.1 | xavier | cw=none**	none	5/5	0.2262	0.0069	[0.2176, 0.2348]	1.1482
59	**sigmoid | sgd | lr=0.01 | he | cw=none**	none	5/5	0.2258	0.0037	[0.2212, 0.2304]	1.0952
60	**sigmoid | sgd | lr=0.01 | xavier | cw=none**	none	5/5	0.2247	0.0035	[0.2204, 0.2290]	1.0968
61	**sigmoid | adam | lr=0.1 | he | cw=balanced**	balanced	5/5	0.2190	0.0374	[0.1726, 0.2655]	1.2714
62	**tanh | adam | lr=0.1 | xavier | cw=balanced**	balanced	5/5	0.2156	0.0179	[0.1934, 0.2379]	1.3977
63	**tanh | adam | lr=0.1 | he | cw=none**	none	5/5	0.2082	0.0076	[0.1988, 0.2177]	1.1548
64	**tanh | adam | lr=0.1 | he | cw=balanced**	balanced	5/5	0.2069	0.0188	[0.1836, 0.2303]	1.3983
65	**relu | adam | lr=0.1 | xavier | cw=none**	none	5/5	0.1923	0.0338	[0.1503, 0.2343]	1.1338
66	**relu | adam | lr=0.1 | he | cw=none**	none	5/5	0.1881	0.0263	[0.1554, 0.2208]	1.1260
67	**sigmoid | sgd | lr=0.001 | he | cw=balanced**	balanced	5/5	0.1736	0.0335	[0.1320, 0.2153]	1.6409
68	**sigmoid | sgd | lr=0.001 | xavier | cw=balanced**	balanced	5/5	0.1717	0.0370	[0.1258, 0.2176]	1.6403

Puesto	Configuración	class_weight	n válidas/total	F1 medio	F1 desvío	IC 95 % de F1	Pérdida val.
69	**sigmoid | sgd | lr=0.001 | xavier | cw=none**	none	5/5	0.1337	0.0261	[0.1013, 0.1661]	1.2099
70	**sigmoid | sgd | lr=0.001 | he | cw=none**	none	5/5	0.1300	0.0225	[0.1021, 0.1579]	1.2133
71	**relu | adam | lr=0.1 | he | cw=balanced**	balanced	5/5	0.0705	0.0191	[0.0467, 0.0942]	1.5184
72	**relu | adam | lr=0.1 | xavier | cw=balanced**	balanced	5/5	0.0703	0.0189	[0.0469, 0.0938]	1.5146

Tabla 1b. Convergencia y costo por configuración, en el mismo orden que la Tabla 1a.

Puesto	Configuración	Mejor época	Épocas ejecutadas	Tiempo hasta la mejor (s)	Tiempo de entrenamiento (s)
1	**tanh | adam | lr=0.01 | he | cw=balanced**	25.8	45.8	2.40	4.21
2	**relu | sgd | lr=0.1 | xavier | cw=balanced**	29.4	49.4	2.04	3.45
3	**tanh | sgd | lr=0.1 | he | cw=balanced**	30.0	50.0	2.07	3.47
4	**tanh | adam | lr=0.001 | he | cw=balanced**	116.6	136.6	10.48	12.34
5	**tanh | adam | lr=0.01 | xavier | cw=balanced**	21.2	41.2	1.91	3.73
6	**relu | adam | lr=0.01 | he | cw=balanced**	19.2	39.2	1.94	3.80
7	**relu | adam | lr=0.001 | xavier | cw=balanced**	58.6	78.6	5.26	7.10
8	**relu | adam | lr=0.001 | he | cw=balanced**	64.0	84.0	5.69	7.49
9	**tanh | sgd | lr=0.01 | he | cw=balanced**	217.0	237.0	14.66	16.06
10	**tanh | adam | lr=0.001 | xavier | cw=balanced**	95.8	115.8	8.61	10.45
11	**tanh | sgd | lr=0.1 | xavier | cw=balanced**	34.2	54.2	2.35	3.75
12	**relu | sgd | lr=0.1 | he | cw=balanced**	27.4	47.4	1.87	3.24
13	**sigmoid | adam | lr=0.01 | xavier | cw=balanced**	37.4	57.4	3.43	5.28
14	**tanh | sgd | lr=0.01 | xavier | cw=balanced**	155.2	175.2	10.52	11.90
15	**sigmoid | sgd | lr=0.1 | xavier | cw=balanced**	50.6	70.6	3.44	4.84
16	**sigmoid | sgd | lr=0.1 | he | cw=balanced**	50.6	70.6	3.46	4.86
17	**sigmoid | adam | lr=0.01 | he | cw=balanced**	28.0	48.0	2.49	4.30
18	**relu | adam | lr=0.01 | xavier | cw=balanced**	13.2	33.2	1.22	3.03

Puesto	Configuración	Mejor época	Épocas ejecutadas	Tiempo hasta la mejor (s)	Tiempo de entrenamiento (s)
19	**tanh | adam | lr=0.001 | xavier | cw=none**	57.6	77.6	5.15	6.94
20	**relu | sgd | lr=0.01 | xavier | cw=balanced**	90.8	110.8	6.14	7.51
21	**tanh | adam | lr=0.001 | he | cw=none**	53.4	73.4	4.73	6.53
22	**relu | sgd | lr=0.01 | he | cw=balanced**	83.4	103.4	5.67	7.06
23	**relu | sgd | lr=0.01 | he | cw=none**	170.6	190.6	11.54	12.90
24	**tanh | adam | lr=0.01 | he | cw=none**	10.4	30.4	0.93	2.74
25	**relu | sgd | lr=0.01 | xavier | cw=none**	171.8	182.6	11.63	12.40
26	**relu | sgd | lr=0.1 | xavier | cw=none**	25.4	45.4	1.73	3.17
27	**relu | adam | lr=0.001 | xavier | cw=none**	49.0	69.0	4.42	6.27
28	**relu | sgd | lr=0.1 | he | cw=none**	22.0	42.0	1.49	2.88
29	**sigmoid | sgd | lr=0.1 | xavier | cw=none**	239.4	259.4	15.91	17.27
30	**sigmoid | sgd | lr=0.1 | he | cw=none**	204.2	221.0	13.65	14.79
31	**tanh | sgd | lr=0.1 | xavier | cw=none**	29.4	49.4	2.01	3.38
32	**tanh | adam | lr=0.01 | xavier | cw=none**	10.0	30.0	0.88	2.68
33	**sigmoid | adam | lr=0.001 | he | cw=balanced**	91.8	111.8	8.14	9.92
34	**sigmoid | adam | lr=0.001 | xavier | cw=none**	258.4	278.4	22.87	24.64
35	**sigmoid | adam | lr=0.01 | he | cw=none**	27.4	47.4	2.42	4.21
36	**sigmoid | adam | lr=0.001 | xavier | cw=balanced**	96.0	116.0	8.50	10.32
37	**sigmoid | adam | lr=0.001 | he | cw=none**	203.6	221.0	17.96	19.51
38	**tanh | sgd | lr=0.1 | he | cw=none**	25.8	45.8	1.75	3.12
39	**relu | adam | lr=0.001 | he | cw=none**	35.0	55.0	3.15	4.93
40	**relu | adam | lr=0.01 | xavier | cw=none**	9.8	29.8	0.86	2.70
41	**relu | adam | lr=0.01 | he | cw=none**	8.2	28.2	0.75	2.59
42	**relu | sgd | lr=0.001 | xavier | cw=balanced**	376.6	392.2	25.52	27.09
43	**tanh | sgd | lr=0.01 | xavier | cw=none**	170.4	188.6	11.49	12.73
44	**tanh | sgd | lr=0.01 | he | cw=none**	142.4	156.0	9.70	10.64

Puesto	Configuración	Mejor época	Épocas ejecutadas	Tiempo hasta la mejor (s)	Tiempo de entrenamiento (s)
45	**sigmoid | adam | lr=0.01 | xavier | cw=none**	27.4	47.4	2.44	4.24
46	**sigmoid | adam | lr=0.1 | he | cw=none**	17.0	37.0	1.51	3.33
47	**relu | sgd | lr=0.001 | xavier | cw=none**	500.0	500.0	35.12	35.61
48	**tanh | sgd | lr=0.001 | xavier | cw=none**	500.0	500.0	34.11	34.17
49	**relu | sgd | lr=0.001 | he | cw=balanced**	328.6	348.6	22.44	23.83
50	**relu | sgd | lr=0.001 | he | cw=none**	500.0	500.0	34.12	34.18
51	**tanh | sgd | lr=0.001 | he | cw=balanced**	443.6	451.6	29.85	30.44
52	**sigmoid | sgd | lr=0.01 | xavier | cw=balanced**	141.0	161.0	9.53	10.91
53	**tanh | sgd | lr=0.001 | xavier | cw=balanced**	314.0	334.0	21.03	22.46
54	**sigmoid | sgd | lr=0.01 | he | cw=balanced**	147.2	167.2	9.85	11.24
55	**sigmoid | adam | lr=0.1 | xavier | cw=balanced**	32.2	52.2	2.90	4.71
56	**tanh | sgd | lr=0.001 | he | cw=none**	500.0	500.0	33.68	33.74
57	**sigmoid | adam | lr=0.1 | xavier | cw=none**	11.6	31.6	1.02	2.83
58	**tanh | adam | lr=0.1 | xavier | cw=none**	17.0	37.0	1.51	3.30
59	**sigmoid | sgd | lr=0.01 | he | cw=none**	498.8	500.0	33.45	33.58
60	**sigmoid | sgd | lr=0.01 | xavier | cw=none**	498.8	500.0	33.55	33.68
61	**sigmoid | adam | lr=0.1 | he | cw=balanced**	22.2	42.2	1.97	3.82
62	**tanh | adam | lr=0.1 | xavier | cw=balanced**	11.2	31.2	1.00	2.83
63	**tanh | adam | lr=0.1 | he | cw=none**	16.0	36.0	1.46	3.29
64	**tanh | adam | lr=0.1 | he | cw=balanced**	21.8	41.8	1.96	3.81
65	**relu | adam | lr=0.1 | xavier | cw=none**	8.6	28.6	0.77	2.60
66	**relu | adam | lr=0.1 | he | cw=none**	9.0	29.0	0.81	2.63
67	**sigmoid | sgd | lr=0.001 | he | cw=balanced**	207.2	227.2	13.94	15.33
68	**sigmoid | sgd | lr=0.001 | xavier | cw=balanced**	196.6	216.4	13.20	14.55
69	**sigmoid | sgd | lr=0.001 | xavier | cw=none**	500.0	500.0	33.55	33.61

Puesto	Configuración	Mejor época	Épocas ejecutadas	Tiempo hasta la mejor (s)	Tiempo de entrenamiento (s)
70	**sigmoid | sgd | lr=0.001 | he | cw=none**	500.0	500.0	33.46	33.52
71	**relu | adam | lr=0.1 | he | cw=balanced**	18.6	38.6	1.67	3.56
72	**relu | adam | lr=0.1 | xavier | cw=balanced**	26.2	46.2	2.37	4.20

2. Comparación visual

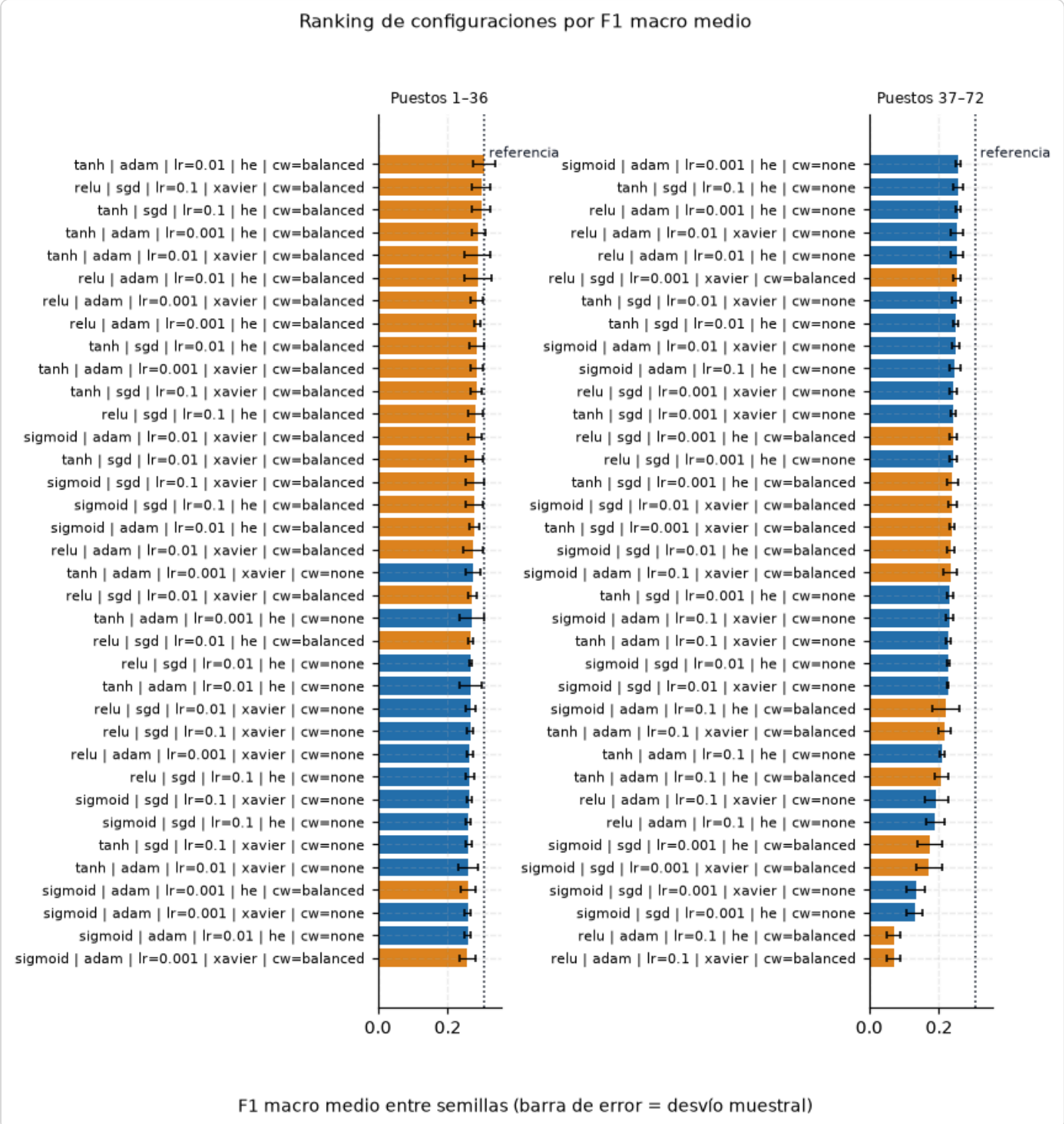


Figura 1 — Ranking de configuraciones. F1 macro medio por configuración a través de sus semillas. Las barras de error son el desvío muestral; la línea punteada marca la mejor media. El color codifica la variante de class_weight (azul = none, naranja = balanced).

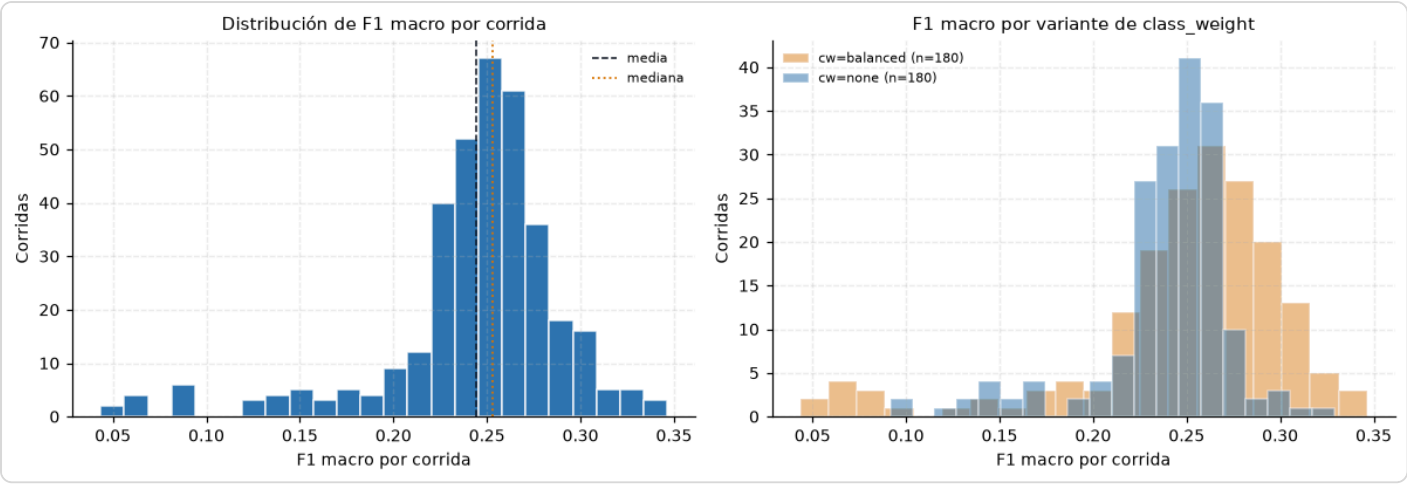


Figura 2 — Distribución de F1. Izquierda: todas las corridas válidas agrupadas. Derecha: las mismas corridas separadas por class_weight. Los histogramas superpuestos muestran si el balanceo desplaza toda la distribución o solo sus colas.

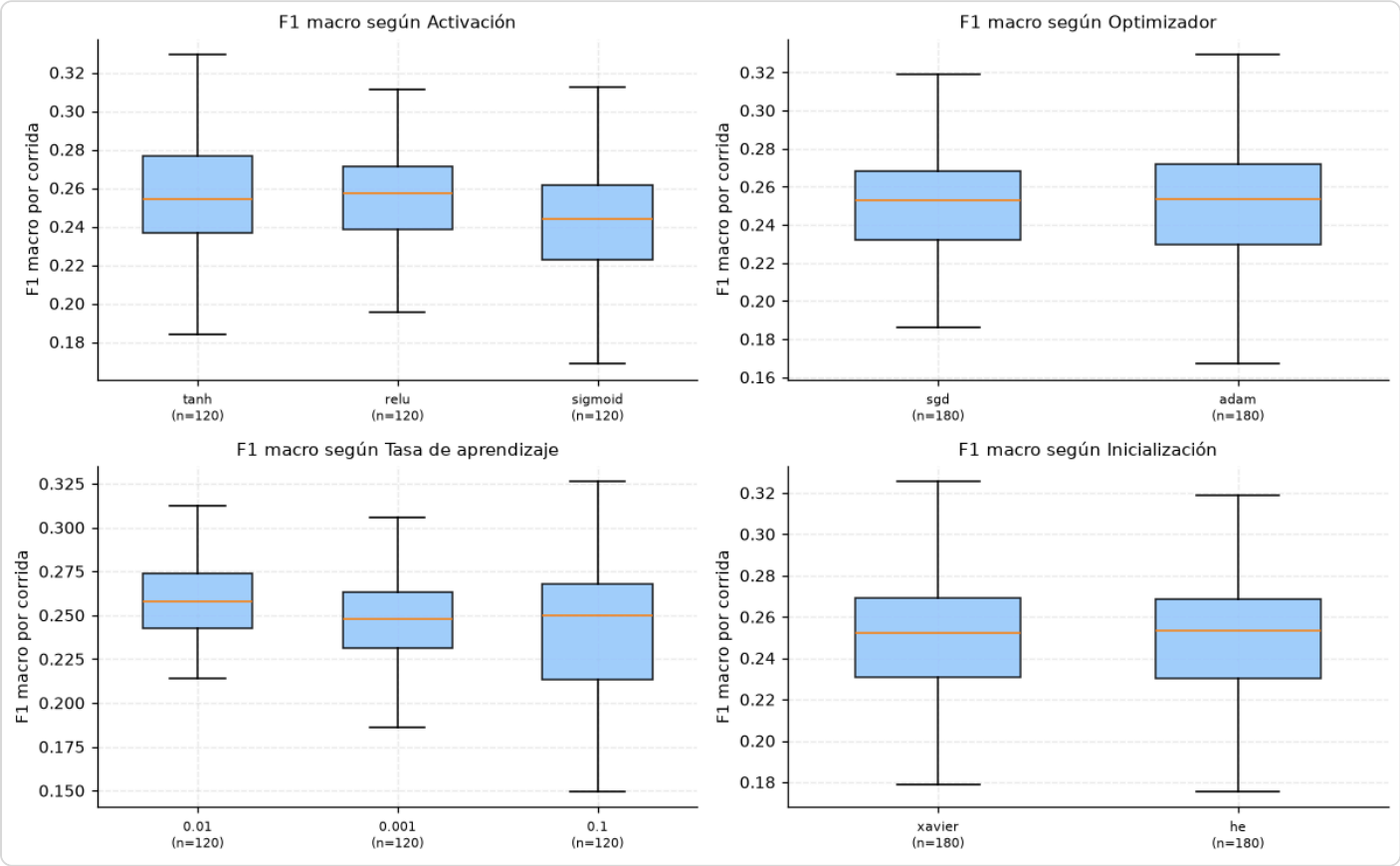


Figura 3 — Boxplots marginales. F1 macro por corrida agrupado por cada dimensión experimental. Son vistas marginales: mezclan configuraciones y semillas, por lo que describen tendencias y no efectos causales aislados.

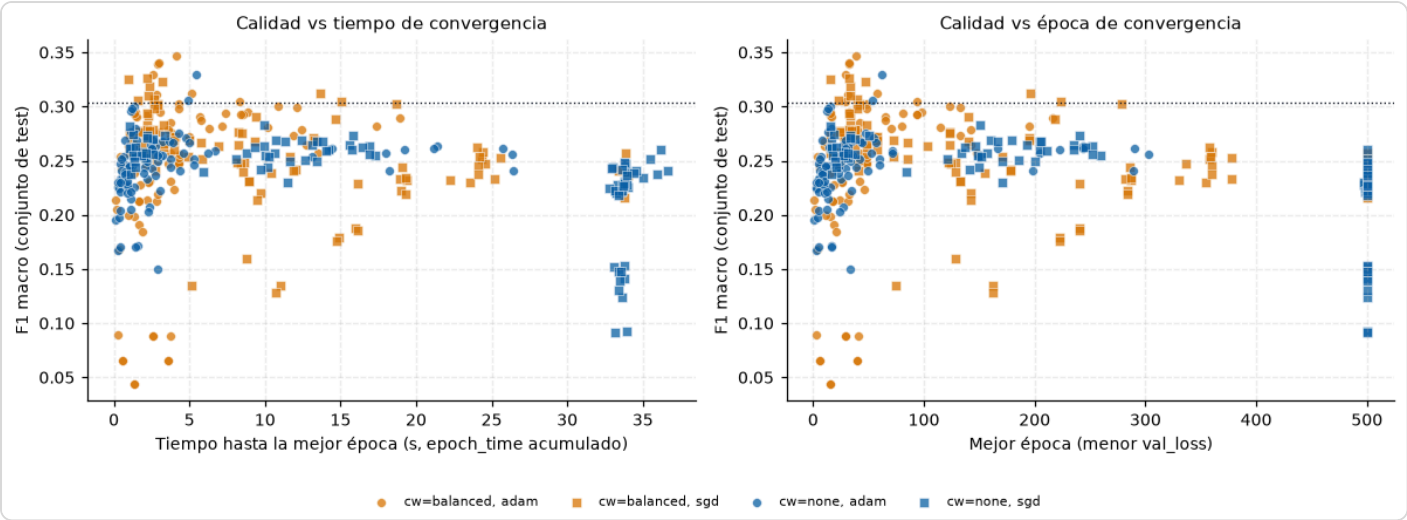


Figura 4 — Calidad vs convergencia. Cada punto es una corrida. Izquierda: F1 contra el tiempo acumulado hasta la mejor época. Derecha: F1 contra la mejor época. El color es class_weight; la forma del marcador es el optimizador.

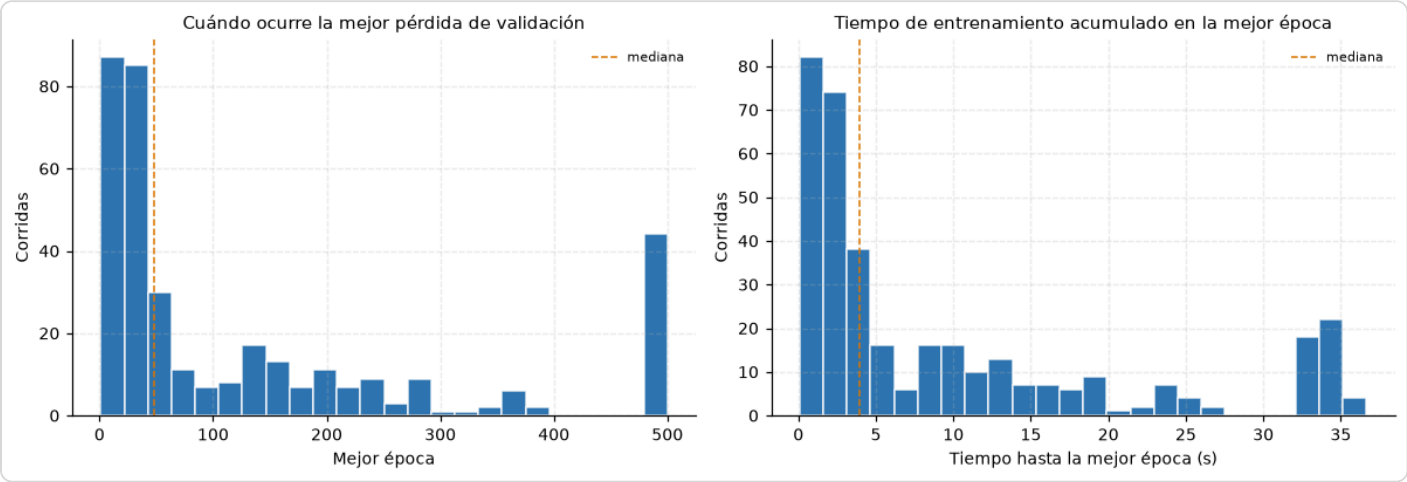


Figura 5 — Histogramas de convergencia. Distribución de la mejor época (menor pérdida de validación) y del tiempo acumulado necesario para alcanzarla. Las colas derechas largas indican configuraciones que siguen mejorando hasta el límite de épocas.

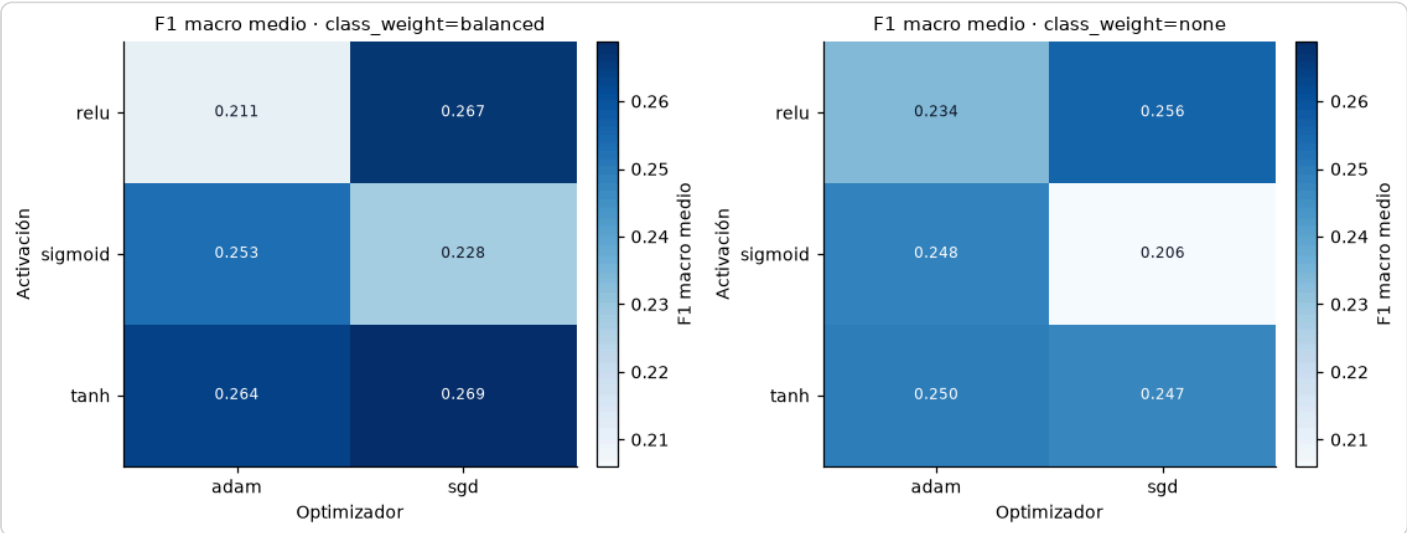


Figura 6 — Heatmap de activación × optimizador. F1 macro medio para cada celda de activación/optimizador, facetado por class_weight. Ambos paneles comparten una escala de color. Las celdas vacías indican que ninguna corrida cubre esa combinación.

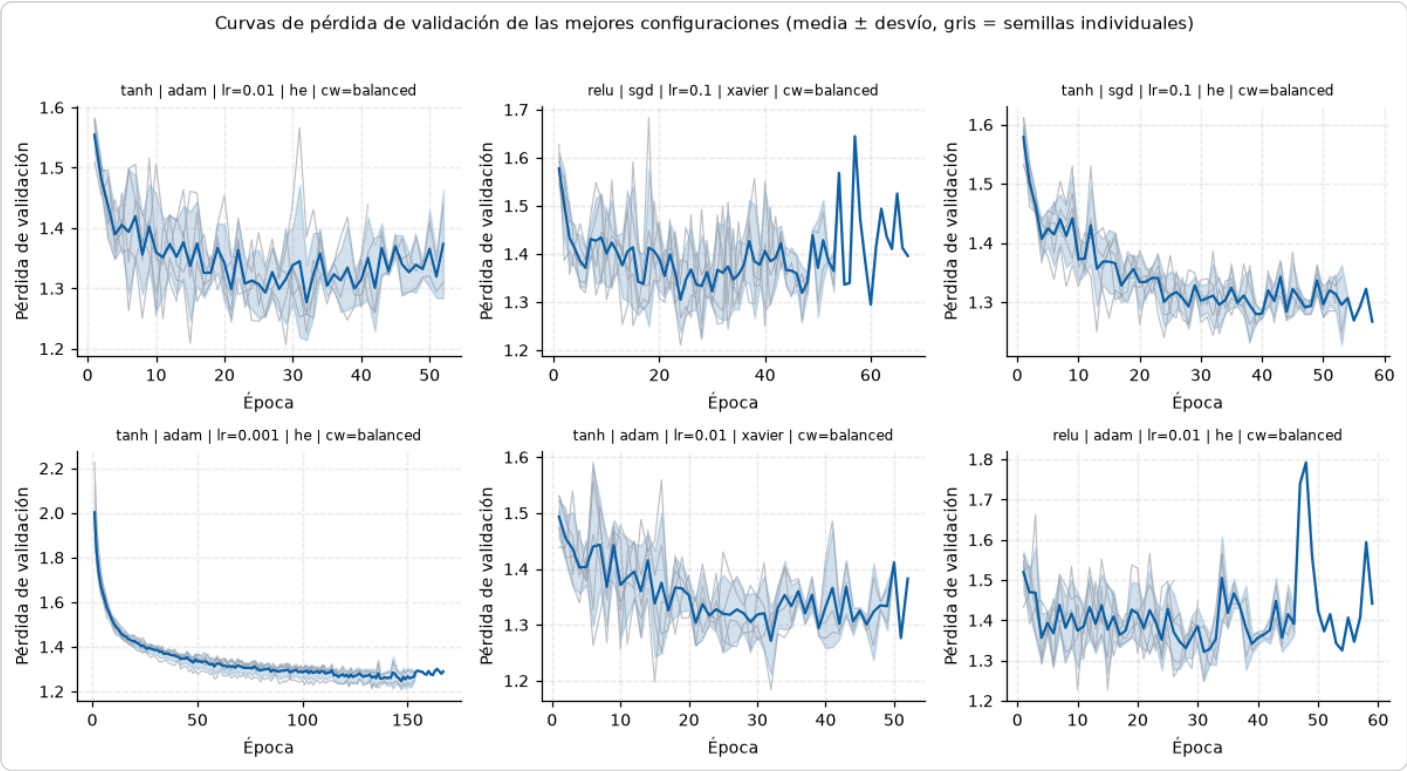


Figura 7 — Curvas de pérdida de validación. Múltiplos pequeños para las mejores configuraciones por F1 medio. La línea azul es la media entre semillas, la banda es ±1 desvío y las líneas grises son las semillas individuales. Las curvas terminan en la época de early stopping de cada corrida.

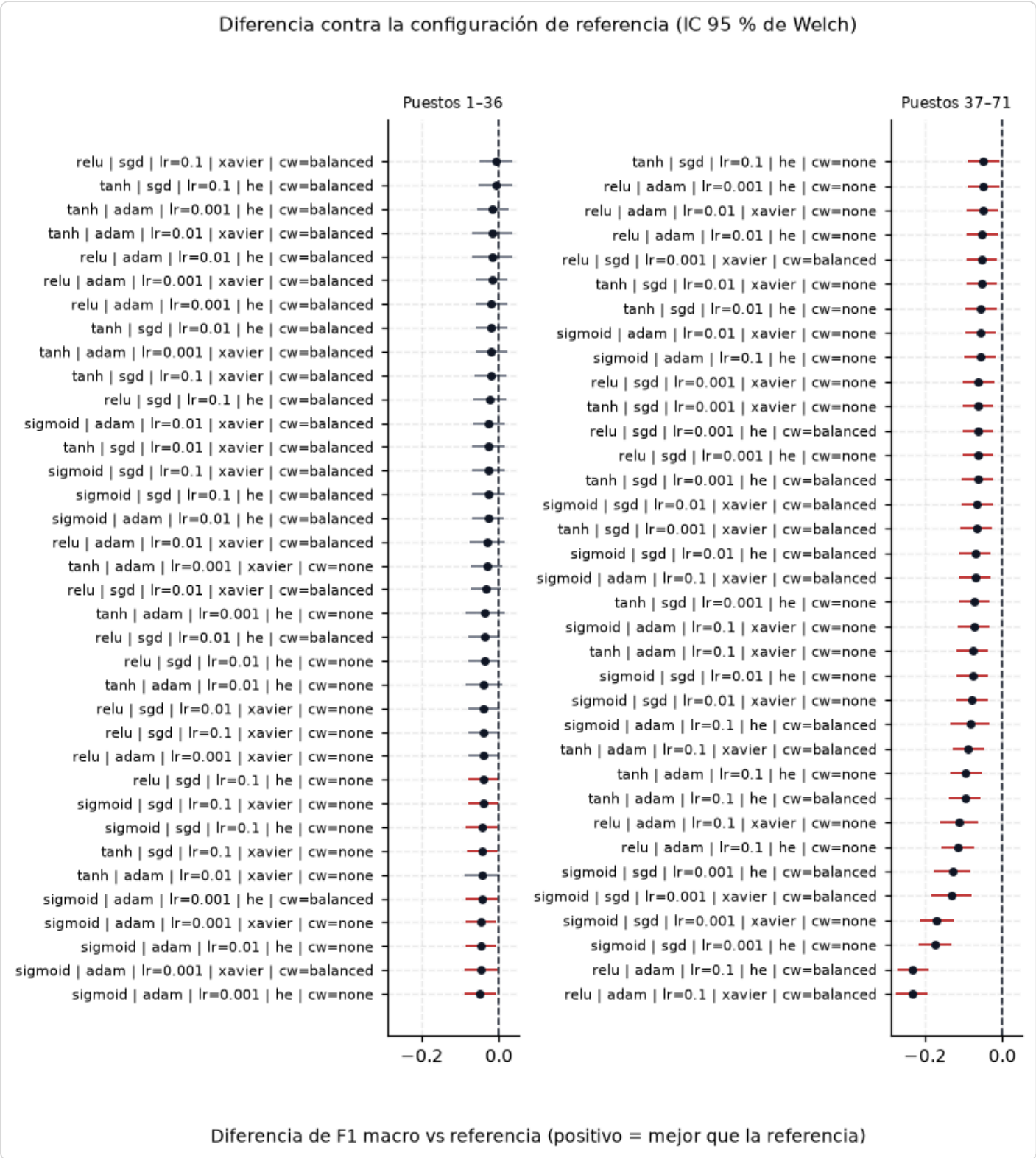


Figura 8 — Forest plot de pares. Diferencia de F1 macro medio contra la referencia, con su intervalo de confianza de Welch al 95 %. Los intervalos que excluyen el cero son los que los datos pueden distinguir de la referencia con este tamaño de muestra.

3. Comparación inferencial contra la referencia

La referencia es la mejor configuración empírica: **tanh | adam | lr=0.01 | he | cw=balanced**. Las diferencias son `config − referencia`; los deltas de F1 positivos son mejores, mientras que los deltas de tiempo y época positivos significan que la configuración es más lenta o más tardía que la referencia.

Tabla 2a. F1 macro contra la configuración de referencia.

Configuración	F1 medio	F1 de referencia	Diferencia F1	IC 95 % de la diferencia
relu | sgd | lr=0.1 | xavier | cw=balanced	0.2963	0.3034	-0.0071	[-0.0507, 0.0364]
tanh | sgd | lr=0.1 | he | cw=balanced	0.2953	0.3034	-0.0081	[-0.0522, 0.0361]
tanh | adam | lr=0.001 | he | cw=balanced	0.2869	0.3034	-0.0164	[-0.0573, 0.0244]
tanh | adam | lr=0.01 | xavier | cw=balanced	0.2854	0.3034	-0.0179	[-0.0691, 0.0332]
relu | adam | lr=0.01 | he | cw=balanced	0.2849	0.3034	-0.0185	[-0.0712, 0.0343]
relu | adam | lr=0.001 | xavier | cw=balanced	0.2844	0.3034	-0.0190	[-0.0596, 0.0216]
relu | adam | lr=0.001 | he | cw=balanced	0.2840	0.3034	-0.0194	[-0.0595, 0.0207]
tanh | sgd | lr=0.01 | he | cw=balanced	0.2839	0.3034	-0.0195	[-0.0612, 0.0221]
tanh | adam | lr=0.001 | xavier | cw=balanced	0.2830	0.3034	-0.0203	[-0.0610, 0.0203]
tanh | sgd | lr=0.1 | xavier | cw=balanced	0.2818	0.3034	-0.0216	[-0.0620, 0.0188]
relu | sgd | lr=0.1 | he | cw=balanced	0.2793	0.3034	-0.0241	[-0.0653, 0.0171]
sigmoid | adam | lr=0.01 | xavier | cw=balanced	0.2778	0.3034	-0.0256	[-0.0669, 0.0157]
tanh | sgd | lr=0.01 | xavier | cw=balanced	0.2770	0.3034	-0.0264	[-0.0693, 0.0165]
sigmoid | sgd | lr=0.1 | xavier | cw=balanced	0.2763	0.3034	-0.0271	[-0.0709, 0.0167]
sigmoid | sgd | lr=0.1 | he | cw=balanced	0.2758	0.3034	-0.0276	[-0.0706, 0.0154]
sigmoid | adam | lr=0.01 | he | cw=balanced	0.2751	0.3034	-0.0283	[-0.0684, 0.0119]
relu | adam | lr=0.01 | xavier | cw=balanced	0.2725	0.3034	-0.0309	[-0.0756, 0.0139]
tanh | adam | lr=0.001 | xavier | cw=none	0.2714	0.3034	-0.0320	[-0.0736, 0.0096]
relu | sgd | lr=0.01 | xavier | cw=balanced	0.2701	0.3034	-0.0333	[-0.0734, 0.0067]
tanh | adam | lr=0.001 | he | cw=none	0.2678	0.3034	-0.0355	[-0.0856, 0.0145]
relu | sgd | lr=0.01 | he | cw=balanced	0.2657	0.3034	-0.0377	[-0.0782, 0.0027]
relu | sgd | lr=0.01 | he | cw=none	0.2649	0.3034	-0.0385	[-0.0790, 0.0020]
tanh | adam | lr=0.01 | he | cw=none	0.2646	0.3034	-0.0388	[-0.0852, 0.0075]
relu | sgd | lr=0.01 | xavier | cw=none	0.2642	0.3034	-0.0392	[-0.0793, 9.0384e-04]
relu | sgd | lr=0.1 | xavier | cw=none	0.2640	0.3034	-0.0394	[-0.0795, 7.1897e-04]
relu | adam | lr=0.001 | xavier | cw=none	0.2634	0.3034	-0.0400	[-0.0802, 2.8301e-04]
relu | sgd | lr=0.1 | he | cw=none	0.2633	0.3034	-0.0401	[-0.0802, -5.7162e-05]
sigmoid | sgd | lr=0.1 | xavier | cw=none	0.2630	0.3034	-0.0404	[-0.0807, -1.4670e-05]
sigmoid | sgd | lr=0.1 | he | cw=none	0.2594	0.3034	-0.0440	[-0.0844, -0.0036]
tanh | sgd | lr=0.1 | xavier | cw=none	0.2594	0.3034	-0.0440	[-0.0843, -0.0038]
tanh | adam | lr=0.01 | xavier | cw=none	0.2590	0.3034	-0.0443	[-0.0893, 6.0247e-04]
sigmoid | adam | lr=0.001 | he | cw=balanced	0.2583	0.3034	-0.0450	[-0.0868, -0.0033]
sigmoid | adam | lr=0.001 | xavier | cw=none	0.2568	0.3034	-0.0466	[-0.0868, -0.0065]
sigmoid | adam | lr=0.01 | he | cw=none	0.2565	0.3034	-0.0469	[-0.0869, -0.0068]
sigmoid | adam | lr=0.001 | xavier | cw=balanced	0.2560	0.3034	-0.0474	[-0.0891, -0.0057]
sigmoid | adam | lr=0.001 | he | cw=none	0.2548	0.3034	-0.0486	[-0.0889, -0.0084]
tanh | sgd | lr=0.1 | he | cw=none	0.2545	0.3034	-0.0488	[-0.0891, -0.0086]
relu | adam | lr=0.001 | he | cw=none	0.2545	0.3034	-0.0489	[-0.0892, -0.0085]
relu | adam | lr=0.01 | xavier | cw=none	0.2520	0.3034	-0.0514	[-0.0919, -0.0110]
relu | adam | lr=0.01 | he | cw=none	0.2513	0.3034	-0.0521	[-0.0925, -0.0117]

Configuración	F1 medio	F1 de referencia	Diferencia F1	IC 95 % de la diferencia
relu | sgd | lr=0.001 | xavier | cw=balanced	0.2503	0.3034	-0.0531	[-0.0932, -0.0130]
tanh | sgd | lr=0.01 | xavier | cw=none	0.2499	0.3034	-0.0535	[-0.0936, -0.0135]
tanh | sgd | lr=0.01 | he | cw=none	0.2481	0.3034	-0.0553	[-0.0956, -0.0150]
sigmoid | adam | lr=0.01 | xavier | cw=none	0.2474	0.3034	-0.0560	[-0.0961, -0.0159]
sigmoid | adam | lr=0.1 | he | cw=none	0.2455	0.3034	-0.0579	[-0.0981, -0.0176]
relu | sgd | lr=0.001 | xavier | cw=none	0.2411	0.3034	-0.0622	[-0.1023, -0.0222]
tanh | sgd | lr=0.001 | xavier | cw=none	0.2405	0.3034	-0.0629	[-0.1032, -0.0227]
relu | sgd | lr=0.001 | he | cw=balanced	0.2402	0.3034	-0.0632	[-0.1033, -0.0231]
relu | sgd | lr=0.001 | he | cw=none	0.2398	0.3034	-0.0636	[-0.1037, -0.0235]
tanh | sgd | lr=0.001 | he | cw=balanced	0.2386	0.3034	-0.0648	[-0.1051, -0.0245]
sigmoid | sgd | lr=0.01 | xavier | cw=balanced	0.2378	0.3034	-0.0656	[-0.1056, -0.0255]
tanh | sgd | lr=0.001 | xavier | cw=balanced	0.2362	0.3034	-0.0672	[-0.1074, -0.0269]
sigmoid | sgd | lr=0.01 | he | cw=balanced	0.2327	0.3034	-0.0707	[-0.1108, -0.0307]
sigmoid | adam | lr=0.1 | xavier | cw=balanced	0.2323	0.3034	-0.0711	[-0.1122, -0.0299]
tanh | sgd | lr=0.001 | he | cw=none	0.2306	0.3034	-0.0728	[-0.1130, -0.0326]
sigmoid | adam | lr=0.1 | xavier | cw=none	0.2291	0.3034	-0.0743	[-0.1144, -0.0342]
tanh | adam | lr=0.1 | xavier | cw=none	0.2262	0.3034	-0.0772	[-0.1175, -0.0369]
sigmoid | sgd | lr=0.01 | he | cw=none	0.2258	0.3034	-0.0776	[-0.1182, -0.0370]
sigmoid | sgd | lr=0.01 | xavier | cw=none	0.2247	0.3034	-0.0787	[-0.1193, -0.0381]
sigmoid | adam | lr=0.1 | he | cw=balanced	0.2190	0.3034	-0.0843	[-0.1358, -0.0329]
tanh | adam | lr=0.1 | xavier | cw=balanced	0.2156	0.3034	-0.0878	[-0.1284, -0.0471]
tanh | adam | lr=0.1 | he | cw=none	0.2082	0.3034	-0.0951	[-0.1354, -0.0549]
tanh | adam | lr=0.1 | he | cw=balanced	0.2069	0.3034	-0.0964	[-0.1373, -0.0556]
relu | adam | lr=0.1 | xavier | cw=none	0.1923	0.3034	-0.1111	[-0.1597, -0.0625]
relu | adam | lr=0.1 | he | cw=none	0.1881	0.3034	-0.1153	[-0.1590, -0.0715]
sigmoid | sgd | lr=0.001 | he | cw=balanced	0.1736	0.3034	-0.1298	[-0.1782, -0.0814]
sigmoid | sgd | lr=0.001 | xavier | cw=balanced	0.1717	0.3034	-0.1317	[-0.1828, -0.0806]
sigmoid | sgd | lr=0.001 | xavier | cw=none	0.1337	0.3034	-0.1697	[-0.2133, -0.1260]
sigmoid | sgd | lr=0.001 | he | cw=none	0.1300	0.3034	-0.1734	[-0.2154, -0.1314]
relu | adam | lr=0.1 | he | cw=balanced	0.0705	0.3034	-0.2329	[-0.2738, -0.1920]
relu | adam | lr=0.1 | xavier | cw=balanced	0.0703	0.3034	-0.2331	[-0.2739, -0.1922]

Tabla 2b. Significancia y tamaño de efecto, en el mismo orden de filas que la Tabla 2a. La última columna marca p de Welch < 0.05; la significancia no mide relevancia práctica.

Configuración	p de Welch	p de Mann-Whitney	delta de Cliff	Efecto	p < 0.05
relu | sgd | lr=0.1 | xavier | cw=balanced	0.714	0.548	-0.280	pequeño	no
tanh | sgd | lr=0.1 | he | cw=balanced	0.683	0.690	-0.200	pequeño	no
tanh | adam | lr=0.001 | he | cw=balanced	0.367	0.421	-0.360	medio	no
tanh | adam | lr=0.01 | xavier | cw=balanced	0.441	0.548	-0.280	pequeño	no
relu | adam | lr=0.01 | he | cw=balanced	0.441	0.548	-0.280	pequeño	no

Configuración	p de Welch	p de Mann-Whitney	delta de Cliff	Efecto	p < 0.05
relu | adam | lr=0.001 | xavier | cw=balanced	0.297	0.310	-0.440	medio	no
relu | adam | lr=0.001 | he | cw=balanced	0.264	0.310	-0.440	medio	no
tanh | sgd | lr=0.01 | he | cw=balanced	0.304	0.421	-0.360	medio	no
tanh | adam | lr=0.001 | xavier | cw=balanced	0.269	0.310	-0.440	medio	no
tanh | sgd | lr=0.1 | xavier | cw=balanced	0.238	0.310	-0.440	medio	no
relu | sgd | lr=0.1 | he | cw=balanced	0.207	0.222	-0.520	grande	no
sigmoid | adam | lr=0.01 | xavier | cw=balanced	0.185	0.310	-0.440	medio	no
tanh | sgd | lr=0.01 | xavier | cw=balanced	0.191	0.222	-0.520	grande	no
sigmoid | sgd | lr=0.1 | xavier | cw=balanced	0.190	0.310	-0.440	medio	no
sigmoid | sgd | lr=0.1 | he | cw=balanced	0.174	0.421	-0.360	medio	no
sigmoid | adam | lr=0.01 | he | cw=balanced	0.133	0.095	-0.680	grande	no
relu | adam | lr=0.01 | xavier | cw=balanced	0.150	0.151	-0.600	grande	no
tanh | adam | lr=0.001 | xavier | cw=none	0.111	0.151	-0.600	grande	no
relu | sgd | lr=0.01 | xavier | cw=balanced	0.086	0.095	-0.680	grande	no
tanh | adam | lr=0.001 | he | cw=none	0.140	0.095	-0.680	grande	no
relu | sgd | lr=0.01 | he | cw=balanced	0.061	0.151	-0.600	grande	no
relu | sgd | lr=0.01 | he | cw=none	0.058	0.095	-0.680	grande	no
tanh | adam | lr=0.01 | he | cw=none	0.089	0.222	-0.520	grande	no
relu | sgd | lr=0.01 | xavier | cw=none	0.054	0.095	-0.680	grande	no
relu | sgd | lr=0.1 | xavier | cw=none	0.053	0.056	-0.760	grande	no
relu | adam | lr=0.001 | xavier | cw=none	0.051	0.095	-0.680	grande	no
relu | sgd | lr=0.1 | he | cw=none	0.050	0.056	-0.760	grande	sí
sigmoid | sgd | lr=0.1 | xavier | cw=none	0.050	0.095	-0.680	grande	sí
sigmoid | sgd | lr=0.1 | he | cw=none	0.039	0.056	-0.760	grande	sí
tanh | sgd | lr=0.1 | xavier | cw=none	0.038	0.016	-0.920	grande	sí
tanh | adam | lr=0.01 | xavier | cw=none	0.052	0.095	-0.680	grande	no
sigmoid | adam | lr=0.001 | he | cw=balanced	0.038	0.032	-0.840	grande	sí
sigmoid | adam | lr=0.001 | xavier | cw=none	0.031	0.056	-0.760	grande	sí
sigmoid | adam | lr=0.01 | he | cw=none	0.030	0.032	-0.840	grande	sí
sigmoid | adam | lr=0.001 | xavier | cw=balanced	0.031	0.032	-0.840	grande	sí
sigmoid | adam | lr=0.001 | he | cw=none	0.027	0.032	-0.840	grande	sí
tanh | sgd | lr=0.1 | he | cw=none	0.025	0.032	-0.840	grande	sí
relu | adam | lr=0.001 | he | cw=none	0.028	0.016	-0.920	grande	sí
relu | adam | lr=0.01 | xavier | cw=none	0.021	0.016	-0.920	grande	sí
relu | adam | lr=0.01 | he | cw=none	0.020	0.032	-0.840	grande	sí

Configuración	p de Welch	p de Mann-Whitney	delta de Cliff	Efecto	p < 0.05
relu | sgd | lr=0.001 | xavier | cw=balanced	0.019	0.016	-0.920	grande	sí
tanh | sgd | lr=0.01 | xavier | cw=none	0.018	0.016	-0.920	grande	sí
tanh | sgd | lr=0.01 | he | cw=none	0.018	0.008	-1.000	grande	sí
sigmoid | adam | lr=0.01 | xavier | cw=none	0.016	0.016	-0.920	grande	sí
sigmoid | adam | lr=0.1 | he | cw=none	0.013	0.032	-0.840	grande	sí
relu | sgd | lr=0.001 | xavier | cw=none	0.010	0.016	-0.920	grande	sí
tanh | sgd | lr=0.001 | xavier | cw=none	0.011	0.008	-1.000	grande	sí
relu | sgd | lr=0.001 | he | cw=balanced	0.010	0.008	-1.000	grande	sí
relu | sgd | lr=0.001 | he | cw=none	0.010	0.008	-1.000	grande	sí
tanh | sgd | lr=0.001 | he | cw=balanced	0.008	0.016	-0.920	grande	sí
sigmoid | sgd | lr=0.01 | xavier | cw=balanced	0.008	0.008	-1.000	grande	sí
tanh | sgd | lr=0.001 | xavier | cw=balanced	0.009	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.01 | he | cw=balanced	0.006	0.008	-1.000	grande	sí
sigmoid | adam | lr=0.1 | xavier | cw=balanced	0.005	0.008	-1.000	grande	sí
tanh | sgd | lr=0.001 | he | cw=none	0.006	0.008	-1.000	grande	sí
sigmoid | adam | lr=0.1 | xavier | cw=none	0.005	0.008	-1.000	grande	sí
tanh | adam | lr=0.1 | xavier | cw=none	0.005	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.01 | he | cw=none	0.006	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.01 | xavier | cw=none	0.006	0.008	-1.000	grande	sí
sigmoid | adam | lr=0.1 | he | cw=balanced	0.005	0.016	-0.920	grande	sí
tanh | adam | lr=0.1 | xavier | cw=balanced	0.002	0.008	-1.000	grande	sí
tanh | adam | lr=0.1 | he | cw=none	0.002	0.008	-1.000	grande	sí
tanh | adam | lr=0.1 | he | cw=balanced	0.001	0.008	-1.000	grande	sí
relu | adam | lr=0.1 | xavier | cw=none	< 0.001	0.008	-1.000	grande	sí
relu | adam | lr=0.1 | he | cw=none	< 0.001	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.001 | he | cw=balanced	< 0.001	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.001 | xavier | cw=balanced	< 0.001	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.001 | xavier | cw=none	< 0.001	0.008	-1.000	grande	sí
sigmoid | sgd | lr=0.001 | he | cw=none	< 0.001	0.008	-1.000	grande	sí
relu | adam | lr=0.1 | he | cw=balanced	< 0.001	0.008	-1.000	grande	sí
relu | adam | lr=0.1 | xavier | cw=balanced	< 0.001	0.012	-1.000	grande	sí

Métricas de convergencia y costo

4. Efectos marginales por dimensión

Cada fila agrupa todas las corridas válidas de un nivel de una dimensión. El p-valor de Kruskal-Wallis evalúa si los niveles de esa dimensión difieren en F1 macro. Como las configuraciones comparten semillas e hiperparámetros, esto es un tamiz descriptivo, no una descomposición factorial.

Tabla 3a. Efecto marginal sobre F1 macro. Kruskal-Wallis se calcula entre los niveles de la misma dimensión.

Dimensión	Nivel	Corridas	Config.	F1 medio	F1 desvío	IC 95 % de F1
Activación	**tanh**	120	24	0.2573	0.0330	[0.2513, 0.2632]
Activación	**relu**	120	24	0.2420	0.0600	[0.2311, 0.2528]
Activación	**sigmoid**	120	24	0.2339	0.0449	[0.2258, 0.2420]
Peso de clase	**balanced**	180	36	0.2486	0.0575	[0.2401, 0.2570]
Peso de clase	**none**	180	36	0.2402	0.0360	[0.2349, 0.2455]
Inicialización	**xavier**	180	36	0.2448	0.0473	[0.2378, 0.2518]
Inicialización	**he**	180	36	0.2439	0.0491	[0.2367, 0.2512]
Tasa de aprendizaje	**0.01**	120	24	0.2614	0.0267	[0.2566, 0.2663]
Tasa de aprendizaje	**0.001**	120	24	0.2395	0.0464	[0.2311, 0.2479]
Tasa de aprendizaje	**0.1**	120	24	0.2322	0.0604	[0.2213, 0.2431]
Optimizador	**sgd**	180	36	0.2455	0.0414	[0.2394, 0.2516]
Optimizador	**adam**	180	36	0.2432	0.0541	[0.2353, 0.2512]

Tabla 3b. Efecto marginal sobre convergencia y costo, en el mismo orden que la Tabla 3a.

Dimensión	Nivel	Tiempo hasta la mejor (s)	Mejor época	p de Kruskal
Activación	**tanh**	8.93	125.8	< 0.001
Activación	**relu**	7.84	110.2	< 0.001
Activación	**sigmoid**	12.19	170.3	< 0.001
Peso de clase	**balanced**	7.49	103.4	< 0.001
Peso de clase	**none**	11.82	167.5	< 0.001
Inicialización	**xavier**	9.67	135.1	0.857
Inicialización	**he**	9.64	135.7	0.857
Tasa de aprendizaje	**0.01**	7.89	113.6	< 0.001
Tasa de aprendizaje	**0.001**	18.12	251.9	< 0.001
Tasa de aprendizaje	**0.1**	2.95	40.8	< 0.001
Optimizador	**sgd**	15.26	225.6	0.876
Optimizador	**adam**	4.04	45.3	0.876

5. Método y definiciones

- **Contrato de entrada.** El reporte lee los CSV `master`, `configs` y `epochs` más recientes del directorio de exportación. Nunca lee el store interno de resultados.

- **Corridas válidas.** Solo las corridas `COMPLETED` y `COMPLETED_WITH_WARNINGS` alimentan las métricas. Las corridas fallidas se cuentan pero se excluyen.
- **class_weight.** Este campo no está en `master`; se une desde `configs` por `config_id`. Un valor faltante significa `none`, nunca un `balanced` implícito.
- **best_epoch.** Época con el mínimo `val_loss`, consistente con `restore_best_weights`.
- **time_to_best_epoch.** Suma acumulada de `epoch_time` hasta `best_epoch`. Excluye el overhead de arranque del proceso, por lo que es comparable entre corridas pero queda levemente por debajo de `training_time`.
- **epochs_executed.** Cantidad de épocas realmente entrenadas antes del early stopping o del tope de épocas.
- **Intervalos de confianza.** Intervalo t del 95 % para la media e intervalo de Welch–Satterthwaite para las diferencias.
- **Tests.** t de Welch (varianzas desiguales) y Mann-Whitney U bilateral. El tamaño de efecto es el delta de Cliff con umbrales de Romano: despreciable < 0.147, pequeño < 0.33, medio < 0.474, grande en el resto.
- **Origen de F1.** `f1_macro` es la métrica final de evaluación registrada en `master` con `evaluation_set = test`. Este módulo no la recalcula.

6. Reproducibilidad

Reconstruí este reporte desde el mismo snapshot con:

```
python -m prebenchmark analyze --experiment-id exp-sweep-01 --export-dir results\exports\exp-sweep-01
```

El reporte es autocontenido: sin acceso a red, sin CSS externo, sin imágenes externas. Las figuras están embebidas como PNG en base64. Los resultados estadísticos dependen solo de los CSV exportados y de la selección de referencia descrita arriba.

Generado por `prebenchmark.analyze` · 360 corridas · 72 configuraciones · referencia `grp-v1-7d12b4d9b658`.